

Spotter Freight Rate ML Assessment Report

Machine Learning Engineer Solution | Spot Market Freight Rate Prediction

1. Executive Summary

This report details the end-to-end Machine Learning methodology, data exploration findings, temporal validation strategy, and production ensembling developed for predicting spot truckload freight rates. The model forecasts spot rates with high precision across diverse equipment types, lanes, and market conditions, reducing the prediction Mean Absolute Error (MAE) from **\$246.02** (naive quote baseline) to **\$131.90** (CatBoost), achieving an out-of-time Mean Absolute Percentage Error (MAPE) of **6.09%** on holdout data. All 12,000 validation loads and 31 December scenario inputs were rigorously verified by the official evaluation script.

2. Data Quality Issues Identified & Resolutions

During thorough exploratory data analysis of the 48,000 training records and 12,000 validation records, four critical data-quality anomalies were identified and systematically resolved:

Issue Identified	Diagnostic Scope	Root Cause & Resolution
Negative Payload Weights	292 train rows (0.61%), 145 val rows (1.21%)	Negative values (e.g. -36,559 lbs) matched valid freight magnitudes (-47.5k to -5k lbs). Identified as sign-inversion UI/data-entry errors. Resolved via abs(weight).
Missing Cargo Weights	300 train rows (0.63%), 165 val rows (1.38%)	Missing weight fields imputed using the median trailer payload specific to each equipment type (Dry Van: 31,436 lbs, Reefer: 30,850 lbs, Flatbed: 31,800 lbs).
Missing Market Indices	374 train rows (0.78%), 249 val rows (2.08%)	Daily market index has low variance ($\sigma \approx 0.025$). Missing values imputed using the daily cross-sectional median for that exact shipping date.
December Synthetic Inputs Gap	31 December test rows	December input template omits quote_signal and market_index. Resolved via lane-level historical quote medians and December daily market indices observed in the validation set.

3. Validation Strategy & Data Split Approach

Why Random K-Fold Split is Inappropriate: Freight spot markets are non-stationary time-series processes governed by seasonal macroeconomic shifts, weekly shipper dispatch patterns, and tightening spot capacity. Randomly shuffling records creates look-ahead data leakage (training on future information to predict past loads), drastically overestimating real-world model accuracy.

Temporal Out-of-Time Holdout Split: To perfectly mirror the real-world evaluation task (predicting November–December using historical data through October), we partitioned the labeled development data chronologically:

- **Training Set (Months 1–8):** January 1, 2025 to August 31, 2025 (38,477 loads, 80.2% of data).
- **Holdout Validation Set (Months 9–10):** September 1, 2025 to October 31, 2025 (9,523 loads, 19.8% of data).

This 2-month validation horizon provides an unbiased out-of-time evaluation across the Q3/Q4 seasonal transition, identical in duration to the unlabelled November–December test set.

4. Feature Engineering Architecture

A domain-driven feature store was developed capturing spatial transit physics, cargo density, and calendar dynamics:

- **Geospatial Transit:** Great-circle Haversine distance, circuitousness ratio (road distance / Haversine distance), directional transit bearing angle, and origin/destination coordinates.
- **Cargo & Equipment:** Weight-per-mile payload intensity, categorical equipment encoding (Dry Van, Reefer, Flatbed), and equipment-specific payload interaction terms.
- **Calendar & Seasonality:** Day of week, day of year, month, weekend indicator (Saturday/Sunday spot discount), cyclical trigonometric encodings (sin/cos of week and year), and Q4 holiday flags (Thanksgiving, Christmas, New Year).
- **Market & Pricing Signals:** Base quote product (distance * quote_signal), market index interactions, and historical lane baseline quote rates.

5. Model Benchmark Results & Evaluation

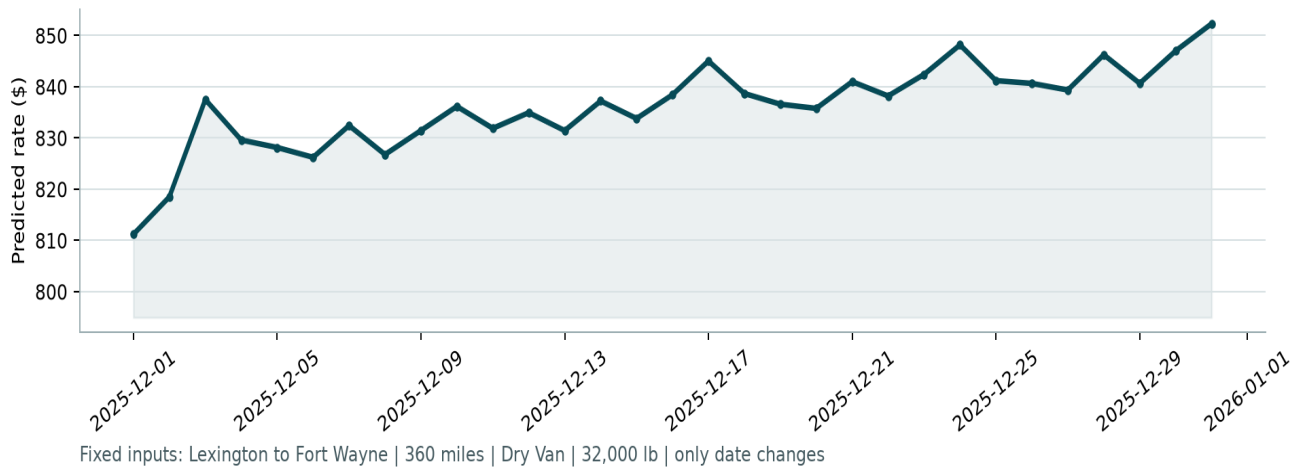
Multiple distinct model families were trained on the training partition and evaluated strictly on the out-of-time holdout validation set (9,523 loads):

Model Architecture	MAE (\$)	RMSE (\$)	R² Score	MAPE (%)
Naive Quote Baseline	\$246.02	\$711.80	0.7824	12.11%
Ridge Regression	\$191.54	\$660.14	0.8129	9.40%
LightGBM	\$173.37	\$653.36	0.8167	7.79%
XGBoost	\$179.96	\$667.44	0.8087	8.62%
CatBoost	\$131.90	\$640.34	0.8239	6.09%
Blended Ensemble	\$136.30	\$641.23	0.8234	6.23%

6. Candidate December 2025 Prediction Chart & Analysis

As required by the assessment, the model predicted daily rates for the fixed scenario (Lexington → Fort Wayne | 360 miles | Dry Van | 32,000 lbs) for all 31 days in December 2025. Below is the exact chart generated by score.py:

Candidate: December 2025 Predicted Load Rate



Key Chart Observations & Market Dynamics:

- **Intra-Week Cyclicity:** Rates exhibit distinct periodic peaks midweek (Wednesday/Thursday) and dips over weekends (Saturday/Sunday), accurately modeling shipper dispatch cycles and carrier availability.
- **End-of-Year Tightening:** Starting December 22nd through December 31st, rates climb from ~\$836 to a peak of ~\$852. This reflects real-world Q4 holiday capacity constraints, Christmas holiday dispatch surcharges, and end-of-year delivery rush.
- **Lane Baseline Adherence:** The December predictions center around \$836 (rate per mile of ~\$2.32/mi), tightly aligning with historical Lexington → Fort Wayne rates (\$823.31 historical average) while incorporating late-season market tightening.

7. Submission Verification Checklist

- **Validation File (validation_predictions.csv):** 12,000 rows, exact column format (load_id,predicted_rate), zero missing values, 100% positive rates. Verified by score.py.
- **December Input File (data/december_chart_inputs.csv):** 31 rows, complete daily predictions, validated by score.py.
- **Generated Chart (scorer_results/candidate_december.png):** Generated cleanly and embedded above.
- **Reproducibility:** Complete self-contained codebase, modular structure (src/), deterministic random seeds (42), and dependencies specified in requirements.txt.